2.4 The universal serial bus (USB) interface

The concept of **USB** can be a little confusing, as it is used to describe several aspects. You may have heard of a USB connection, a USB cable, or maybe you call a storage device 'a USB'. You need to understand the difference between several of these aspects. A USB interface includes items such as a **USB port**, a **USB cable**, a **USB connection** and a **USB device**.

A USB device, such as a USB memory stick, is a device that uses the USB interface. It plugs into a USB port on a computer and it transmits data using a USB connection. A USB cable can be used to connect devices, such as a mouse, to a computer.

A USB connection uses serial data transmission to transmit data between the devices. However, it a special type of serial data transmission connection that is used that is designed to transmit data at high speeds.

ACTIVITY 2.5

One of the issues with serial data transmission is the transmission speed is slower than parallel. Research how a USB connection is able to transmit data at a faster rate.

COMPUTER SCIENCE IN CONTEXT

Many devices are connected using a USB cable. This is because the USB interface is an industry standard, which means that manufacturers don't need to worry about compatibility issues when connecting devices. One of the most recent versions of the USB interface to be released is USB 4.

ACTIVITY 2.6

Research the data transfer speed of USB 4; is this different to USB 1, 2 and 3? What other differences can you find between the different versions of USB?

KEY WORDS

USB: an industry standard that is used to transmit data.

USB port: a socket that is a part of a device or computer that enables you to insert a USB cable.

USB cable: a type of transmission media that uses the USB method to transmit data.

USB connection: a collective name for using a USB cable plugged into a USB port to transfer data from one device to another.

USB device: the name of a device that plugs into a USB port on a computer.

ethernet: another type of connection that can be used to transmit data within a network.

There are several advantages and disadvantages of the USB interface, see Table 2.3:

Advantages	Disadvantages
It is a simple interface. The USB cable to device can only fit into the USB one way. Therefore, it means less errors in connecting devices are likely to be made.	The length of a USB cable is limited, normally to 5 metres.
The speed of a USB connection is relatively high, so data can be transferred quickly.	The transmission speed is relatively high for a USB connection, but it isn't as high as other types of connection, such as **ethernet**.

(continued)

Advantages	Disadvantages
The USB interface is a universal standard for connecting devices, therefore a USB port is included in many different devices.	
When a USB device to cable is inserted into a USB port, it is automatically detected. The first connection will normally involve the download of drivers to operate the hardware that has been connected. Each connection after this the driver should not need to be downloaded again.	
A USB connection can also be used to power a device, so it does not need another power source. It can also use this to charge a device, such as a mobile phone.	

Table 2.3: Advantages/disadvantages of the USB interface

Questions

12 What does USB stand for?

13 What type of data transmission method does a USB interface use?

14 What are **two** advantages of using the USB interface?

15 Can you name **five** devices that are connected using a USB interface?

16 What is the length a USB cable limited to? Can you think why it is limited to this length?

2.5 Detecting errors in data transmission

The transmission of data from one device to another is often not a perfect process. In the process of transmitting the data, interference can occur. This can cause data to be lost, data to be gained and data to change. This would not be helpful and could also lead to many issues. If a password wasn't transmitted correctly, then that person may not be able to log into their account. If a person's address wasn't transmitted correctly, they may not get email that needs to be sent to them. If a person's details are not transmitted correctly to a company, they may not get a product or service that they order. The accuracy of data is often vital, therefore there needs to be a procedure in place to detect any errors in data so that actions can be taken to correct this.

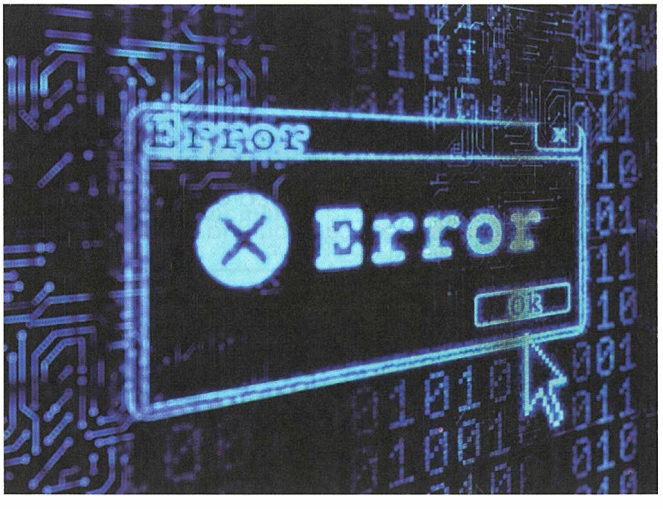


Figure 2.10: Procedures need to be in place to handle errors in data transmission